# SUPERVISED LEARNING

## High-dimensional Features

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$$

- $x_1$: Fasting Blood Sugar (mg/dL)
- $x_2$: Body Mass Index (BMI)
- $x_3$: Age (years)
- $x_n$: Average Daily Physical Activity

$$y = 0/1$$ Developed Diabetes

Lecture 5? **Feature Engineering:** the process of selecting and transforming data attributes to enhance the performance of machine learning models.

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## What Kind of Labels?

The dataset is a collection of tuples: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ Where:

- $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_m^{(i)}]$ is a vector of features for the $i^{th}$ instance.
  - $y^{(i)}$ is the continuous target variable for the $i^{th}$ instance [Regression].
  - $y^{(i)}$ is the categorical (discrete) class label for the $i^{th}$ instance [Classification].